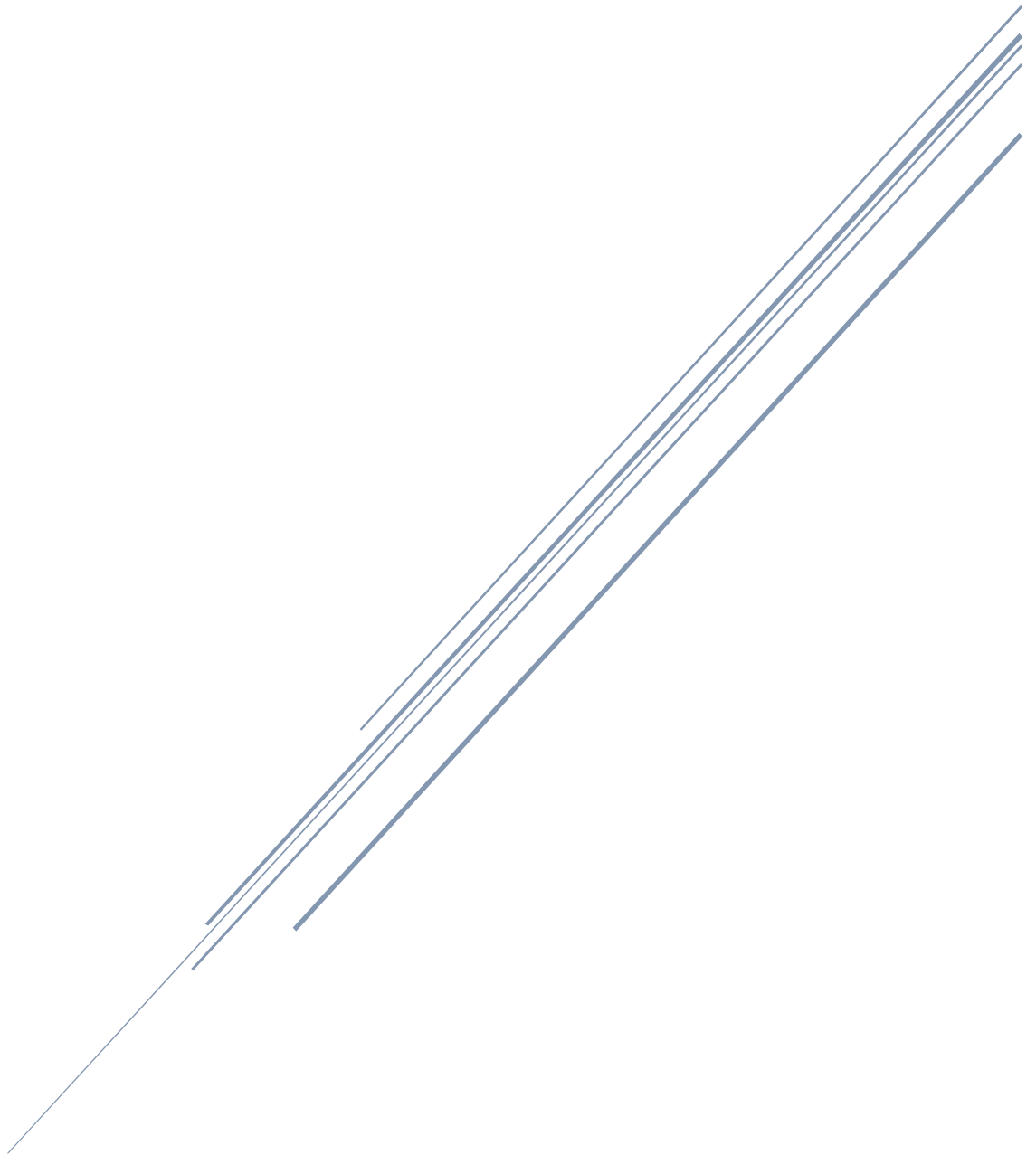


ASSIGNMENT 1

Human Computer Interaction Design



Makaya
Guzu

Introduction

In the evolving cyberspace, IT learners require engaging platforms that go beyond traditional classroom methods. A comprehensive website has been developed to empower IT students by offering real-time access to scholarly materials and peer mentoring opportunities. This platform will support students through three key features.

First, an interactive quiz module enables self-assessment and immediate feedback on various topics, helping students identify weaknesses and reinforce their learning. Second, it provides tailored support packages for creating Portfolios of Evidence (POEs) with the necessary guidance, templates, and resources essential for professional documentation.

Third, an online booking system simplifies scheduling for workshops, group studies, or individual tutorials, reducing administrative burdens and improving accessibility to needed assistance. The document will detail the site's proposed functionality, focusing on interaction design principles like consistency, learnability, and feedback. By targeting usability goals such as effectiveness, efficiency, and user satisfaction, the end product aims to be a functional, interactive learning environment that enhances academic success for all IT students.

1.1 Detailed Functionality of the Website

My website presents new visitors with a straightforward homepage featuring two distinct paths to choose from: Register or Log in. New visitors need to choose Register to submit their student number along with their email address and password. After successful confirmation of their information the system sends a welcome email containing a verification link. After their verification process users can access their accounts using their username along with their password. Users who have previously registered can log in through direct access.

After logging in to their accounts students access their personal dashboard which serves as the central control platform of the website. Students find three primary sections on their dashboard along with a side menu to enter the Discussion Forum: Quizzes, POE Support and Class Booking. Students can track their academic progress through the dashboard which displays both reminders and a progress tracker to provide an overview of their current academic standing.

A student who wants to take a quiz needs to click on Quizzes. The system shows them various filtering options which let them choose content according to specific subjects like Networking Basics and different modules and difficulty ratings. The quiz starts after the user makes their selection. The assessment contains three types of questions which consist of multiple-choice items and true/false statements and interactive coding exercises. The system shows students immediate feedback by displaying explanations about their correct and incorrect answers. The results page will show their score together with a performance chart at the conclusion. The assessment confirms the completion of the process together with showing areas which need improvement.

Students who wish to work on their Portfolio of Evidence (POE) need to select the POE Support option. The platform delivers step-by-step guidance through downloadable templates and video tutorials which serve as instructional materials. The checklist helps them verify that all requirements have been fulfilled. The system shows a "Submission Successful" message after users complete their POE upload through drag-and-drop functionality.

The student needs to choose Book a Class for academic assistance. The system displays a

calendar which shows all available appointment times. The student needs to select an available slot and provide their consultation topic before selecting Confirm Booking. A confirmation message will appear on the screen while the system sends an automatic email alert. The dashboard will display a new appointment reminder.

The student needs to use the Discussion Forum when they want to get help from their peers. Users have the option to explore existing discussions while they can also find solutions through search or create their own questions. The system displays notifications to users when they receive replies from their lecturers or peers.

The student will access their dashboard after they complete their main responsibilities which include finishing a quiz or uploading a POE or booking a consultation they can choose to continue with another task or log out. The system displays a logout confirmation message which directs users back to the homepage.

The website provides users with a logical path which delivers precise assistance at every step from initial registration to final task completion.

1.2.1 Usability Goals

Sharp, Rogers & Prece (2019) stated that usability is the process of making interactive goods simple to understand, efficient to use, and pleasurable for the user. It entails making the most of people's interactions with interactive products so they can perform their daily tasks at work, school, and other places (Sharp, Rogers & Preece, 2019).

Effectiveness

I will achieve effectiveness through the system by delivering specific results for each task that I finish. When I open the quiz section and choose Networking Basics the system shows me whether my answers are right or wrong immediately after each question. The evaluation will present me with a performance chart and score that demonstrate my learning achievements together with my strong points and areas that need improvement. The method will help me establish a better study system which will reveal any missing information at an early stage.

Efficiency

The system design will focus on reducing the total number of steps which users need to finish their tasks. I will use the Book a Session button to schedule a tutorial by selecting a free time slot in the calendar before confirming my appointment. The process will require two mouse clicks before the confirmation message appears. By keeping this process simple, I will save time and be able to focus more on studying instead of struggling with complicated navigation.

Safety

The protection of academic data and error prevention will receive my complete focus to maintain safety standards. The system will verify my login information through username and password verification while protecting my personal details from unauthorized access. The system will alert me when I enter wrong login information but it will keep my personal data safe from exposure. I will stop students from making duplicate appointments by blocking off time slots that have already been booked in the booking system. My data remains protected through this method while my schedule stays up to date.

Utility

I will create the website to focus on usefulness by adding features which serve my needs as a student. The support section will help me during my Portfolio of Evidence (POE) work because I can download templates and follow the detailed instructions. The ready-made resources will help me complete my coursework directly so I don't have to search for additional help.

Learnability

The platform will receive my attention to create an interface which assists new users in learning the system. The system will show me a brief tutorial after my initial login which demonstrates how to find quizzes and POE support and class booking options. The interface will become user-friendly through my implementation of particular icons and precise labels which will eliminate the need for additional training.

Memorability

I will maintain design consistency to help users remember information better. My dashboard layout together with the sidebar menu will appear in the same place every time I log in so I can easily identify them. The system will remain familiar so I can find what I need without any problems after my break from using it.

1.2.2 Desirable Aspects of User Experience

Satzinger, Jakson, and Burd (2016) argued that A customer's or user's whole experience dealing with a system, product, or service is referred to as user experience, or UX for short. Ease of use, accessibility, visual design, user interface features, and the emotional impact of using the product are all factors that UX considers when evaluating a product or service (Sharp, Rogers & Preece,2019).

Satisfying

I will present the outcomes of my work on the website to create user satisfaction. Users will receive their test results immediately after quiz completion because they will get their scores and answer explanations. The automatic updates from my progress tracker will reveal my progress toward achieving my academic targets. The straightforward feedback will help me maintain motivation while keeping my learning confidence intact.

Enjoyable

The user interface will become enjoyable through my work to create a simple design which remains easy to use. The system will show me a dashboard screen after I enter my account where I can access Quizzes and POEs and Class Bookings. The website navigation system lets me reach any section through a single mouse click. The basic design will help me stay at ease while motivating me to use the platform regularly.

Engaging

The system will generate user engagement through its reward program and achievement tracking system. The system awards points to users who complete cybersecurity quizzes which enables them to obtain badges like Network Explorer. I will be able to check my position on the leader board. The gamified elements will help me maintain enthusiasm for ongoing learning while driving me to compete with my peers.

Pleasurable

The system will receive small motivational elements which will improve user satisfaction. When I answer correctly, I will see a green tick and a message saying "Well done!". The platform shows digital confetti as a celebration when users finish their quizzes. Small thoughtful details in the

system will show me appreciation which will build a positive connection between me and the system.

Exciting

The experience will stay exciting through my plan to introduce fresh challenges and new content on a regular basis. The system will award me with a bonus quiz and special badge when I finish five quizzes during one week. The platform will keep users coming back through its steady stream of new content which maintains their interest.

Entertaining

My plan involves creating entertaining features which will make learning into a pleasurable activity. The POE builder enables me to organize sections through a drag-and-drop system which replaces the need for traditional form entry. I can now start timed quiz competitions with my friends which makes studying feel more like playing a game. The educational tools will help students complete their academic work through enjoyable interactive methods.

1.2.3 Design Principles

Sharp, Rogers & Prece (2019) argued that Design Principles Design principles are used by interaction designers to aid their thinking when designing for the user experience (Sharp, Rogers & Prece ,2019). These are generalizable abstractions intended to orient designers toward thinking about different aspects of their designs.

Consistency

My ability to adapt to the system will become easier because of my consistent approach to system interaction. The Take Quiz button maintains its exact location and design throughout the system. The consistent button placement helps me save time because I do not need to search for it each time I want to start a quiz.

Visibility

Visibility stands as my top priority since I will showcase essential features through prominent display. Upon logging into the system I will find Book Class and View Resources displayed prominently on my dashboard. The system's straightforward display layout will help me navigate without wasting time on unnecessary searches.

Feedback

My actions will receive instant feedback which will provide reassurance about their success. Immediately after submitting a quiz I will receive my results together with complete explanations. After booking a class the system will display an instant confirmation message and send an email notification which verifies my reservation has been successfully recorded.

Constraints

The system will apply constraints to stop users from making errors. The booking platform will show only free time slots which stops users from scheduling multiple appointments at the same time. The system requires students to answer every question before submitting a quiz to prevent both errors and partial submissions.

Affordance

The platform design will incorporate affordance principles which make interactive components naturally understandable. The buttons will display obvious click ability and calendars will display

scheduling options visually while book and calendar icons will use their design to show their functions.

The application of these design principles will create a website that users find straightforward to use and dependable and easy to navigate. The platform will enable students to finish their work quickly through a simple academic support network.

1.2.4 Interaction Types

Instructing

I will use instructing as one of the essential ways to interact with the system. The system requires me to start the selection process for quizzes and then begin each quiz and submit answers and book online classes and download POE resource packages. The direct instructions will enable me to reach my objectives at a fast pace while avoiding any unnecessary procedures.

Conversing

The system will accept my input through dialogue structures which I will use to communicate with it. The Chabot will receive my questions about locating particular POE guides and changing my class schedule. I will join the discussion forum to work with fellow students and teachers who will help me find solutions to academic problems and share learning materials. The platform will develop a social atmosphere through this conversational approach which will make team communication easier.

Manipulating

The user will maintain control of digital elements through manipulation to perform actions. The system requires me to upload my POE material by using drag-and-drop features and I need to solve coding problems that appear in the quiz section and organize my learning resources on the dashboard. The learning process will become more interactive and useful through this method of communication.

Exploring

I will search through the platform to discover resources which I can find on my own. The search functions and tutorial filters and IT modules will help me find materials that match my educational requirements. By selecting topics that interest me I will control my educational path through exploration.

Responding

The system will deliver answers which match my individual requirements so I will gain advantages from this. The system will offer alternative practice questions and extra lecture materials and POE support packages when I struggle with database quizzes. The registration process will send me notifications about upcoming deadlines for my classes. The system will deliver personalized support which matches my academic targets and educational progress.

1.2.5 Social Interactions

Interaction design emphasizes the importance of **social interaction** in digital platforms, as learning is often enhanced through collaboration and communication (Sharp, Rogers & Preece, 2019).

Peer Discussion Forum

I will participate in the peer discussion forum to connect with other students who will help me solve questions and share resources and exchange intellectual ideas. Students who have dealt with similar problems will assist me through forum posts when I encounter coding obstacles and module clarification needs. I will share my personal test preparation strategies and important resources with this group. The program will help me build relationships with other learners who work together like IT professionals do in actual work environments.

Lecture-Student Consultations

The direct academic support will become available to me through my participation in lecture-student consultation sessions. The system allows me to schedule appointments with lecturers and tutors for both individual and group learning sessions. I will enter the video call at the right moment to present my problems so I can get specific advice. The digital communication system will help me get academic assistance while keeping my relationships with teachers strong.

The combination of peer discussions with lecturer consultations will deliver dual benefits of group-based learning and customized academic assistance. The social interactions will enhance my educational results and develop my abilities to work in professional IT teams.

1.2.6 Emotional Interaction

Sharp, Rogers & Preece (2019) argued that emotional interaction emphasizes creating interfaces that are not only functional but also enjoyable, motivating, and supportive of the user's emotional state throughout their journey.

Peer Discussion Forum

I will create emotional ties to the platform through the peer discussion forum by participating in dialogues with my fellow students. The forum will be my tool to get help from fellow students who understand my coding problems and module questions. I will distribute my POE preparation techniques together with useful materials to achieve a sense of contribution. The collaborative environment will link me to a team which shows how IT professionals work together and this will boost my confidence and study motivation.

Lecture-Student Consultations

The meetings between lecturers and students will help me find comfort and drive in my educational journey. I will book single or multiple appointments to get video access for speaking with lecturers and tutors who will provide customized help with my problems. Online communication enables me to maintain academic relationships with my peers and faculty members which in turn provides me with both understanding and support.

Gamification and Motivation

The system will establish an emotional bond with me through its gamification elements. The system will show my progress through badges and points and achievement levels which I will receive as rewards. The recognition I receive will build my confidence while it keeps me motivated to visit regularly. The visible progress indicators will show me exact signs of my development which will help me stay focused on my ongoing improvement.

Interface Design and Emotional Support

The design will provide me with emotional support through its combination of warm colours and contemporary visuals and straightforward layouts and optimistic language. The system interface design choices will decrease user stress when taking coding quizzes while making the platform

user-friendly instead of complex. The design shows careful consideration which helps me feel comfortable and motivated during my educational process.

The platform will maintain my long-term engagement and motivation through its mixture of collaborative features and academic support and gamification elements and emotionally supportive design.

1.2.7 Web Content Accessibility Guidelines (WCAG) 2.0

Sharp, Rogers & Preece (2019) argued that accessibility is a central concern in interaction design, ensuring that digital systems can be used by as many people as possible, including those with disabilities. The IT Student Support Website will follow the **Web Content Accessibility Guidelines (WCAG) 2.0**, which are built around four key principles: perceivable, operable, understandable, and robust.

Perceivable: Every visual element on the website will feature appropriate alternatives. Video resources must have captions to assist students with hearing impairments while images require descriptive text (alt-text). The website will enhance its readability for visually impaired users by offering high-contrast colours and adjustable font sizes.

Operable: Students who use keyboard navigation instead of a mouse will be able to complete tasks like class booking and quiz initiation through full website support. The website will employ uniform menus and navigation structures to decrease the amount of work required from users.

Understandable: The instructions will use basic language without complicated technical terms except when absolutely required. Students with different abilities will understand and utilize the POE guides because the content contains detailed step-by-step instructions.

Robust: The platform will undergo compatibility testing with assistive technologies including screen readers and voice input systems to guarantee all students with diverse needs can access every feature.

The IT Student Support Website implementation of WCAG 2.0 principles will establish accessibility standards and create an environment which provides equal learning opportunities for all students to engage and succeed.

REFERENCE LIST

Satzinger, J., Jackson, R., and Burd, S., 2016. *Systems Analysis and design in a Changing World*. 7th; Boston; Cengage Learning.

Sharo, H., Rogers,. Y., and Preece, J., 2019. *Interaction Design: Beyond Human-Computer Interaction*. 5th Edition; Indiana; Wiley.

Appendix A - PoE Marking Rubrics

MODULE NAME:	MODULE CODE:
HUMAN COMPUTER INTERACTION	HCIN6222/p/w

STUDENT NAME:	Makaya Guzu
STUDENT NUMBER:	ST10404851

PART 1 RUBRIC – SKELETON OUTLINE					
Criteria	LEVELS OF ACHIEVEMENT				Feedback
	EXCELLENT	GOOD	DEVELOPING	POOR	
	Score Ranges Per Level (1/2 marks possible)				
GENERAL:	8-10	7-6	5	0-4	
Language usage, referencing, and editing of the document	Language usage, referencing, and editing of the document are excellent.	Language usage, referencing, and editing of the document are good.	Language usage, referencing, and editing of the document are satisfactory.	Language usage, referencing, and document editing are poor or completely unsatisfactory.	
Description of website functionality	16-20	10-15	6-9	0-5	
	An excellent description that clearly describes the website’s functionality is included.	A good description that adequately describes the website’s functionality is included.	A description that is slightly vague in describing the functionality that the website will offer is included.	A very unclear description of the functionality offered by the website, or the description is missing.	

Usability goals	9-10	6-8	4-5	0-3	
	An excellent description of how usability goals will be implemented in the website's design is included.	A good description of how usability goals will be implemented in the website's design is included.	Some details regarding how usability goals will be implemented in the website's design are included.	The description regarding how usability goals will be implemented in the website's design is not included or is completely unsatisfactory.	
Desirable aspects of user experience	9-10	6-8	4-5	0-3	
	An excellent description of how desirable aspects of user experience will be implemented in the website's design is included.	A good description of how desirable aspects of user experience will be implemented in the website's design is included.	Some details regarding how desirable aspects of user experience will be implemented in the website's design are included.	The description regarding how desirable aspects of user experience will be implemented in the website's design is not included or is completely unsatisfactory.	
Design principles	9-10	6-8	4-5	0-3	
	An excellent description of how design principles will be implemented in the website's design is included.	A good description of how design principles will be implemented in the website's design is included.	Some details of how design principles will be implemented in the website's design are included.	The description regarding how design principles will be implemented in the website's design is not included or is completely unsatisfactory.	
Interaction types	9-10	6-8	4-5	0-3	
	An excellent description of how interaction types will be implemented in the website's design is included.	A good description of how interaction types will be implemented in the website's design is included.	Some details regarding how interaction types will be implemented in the website's design are included.	The description of how interaction types will be implemented in the website's design is not included or is unsatisfactory.	

Social Interactions	9-10	6-8	4-5	0-3	
	An excellent description of how social interactions will be implemented in the website's design is included.	A good description of how social interactions will be implemented in the website's design is included.	Some details of how social interactions will be implemented in the website's design are included.	The description regarding how social interactions will be implemented in the website's design is not included or is completely unsatisfactory.	
Emotional interaction	9-10	6-8	4-5	0-3	
	An excellent description of how emotional interactions will be implemented in the website's design is included.	A good description of how emotional interactions will be implemented in the website's design is included.	Some details regarding how emotional interactions will be implemented in the website's design are included.	The description regarding how emotional interactions will be implemented in the website's design is not included or is completely unsatisfactory.	
Web Content Accessibility and Guidelines (WCAG) 2.0	9-10	7-8	4-5	0-3	
	An excellent description of how WCAG will be implemented in the website's design is included.	A good description of how WCAG will be implemented in the website's design is included.	Some details regarding how WCAG will be implemented in the website's design are included.	The description regarding how WCAG will be implemented in the website's design is not included or is completely unsatisfactory.	
Part 1 TOTAL					/100